

**AUTONOMOUS
PARALLEL
PARKING SYSTEM**

Abstract:

This project presents the design and implementation of an autonomous parallel parking system uses ultrasonic sensors to detect obstacles and measures available parking space. Based on the sensor data the controller automatically performs parking without human intervention. A servo motor is used for steering control and an LCD display shows real-time status such as searching, parking and parked. This system improves accuracy, safety and convenience.

Introduction:

Parallel parking is one of the most difficult driving tasks, especially in congested areas. It requires precise control and judgment. To overcome this difficulty, an autonomous parking system is developed. This system uses sensors and a microcontroller to detect space and

park the vehicle automatically. It is a basic concept used in modern smart vehicles and driver assistance systems.

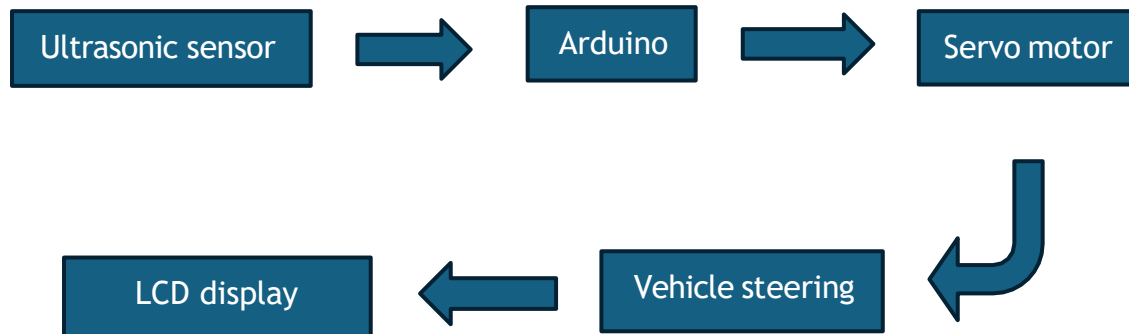
Objective:

1. To design an autonomous parking system.
2. To detect obstacles using ultrasonic sensors.
3. To control steering using servo motor.
4. To display system status using LCD.
5. To reduce human effort and improve safety.

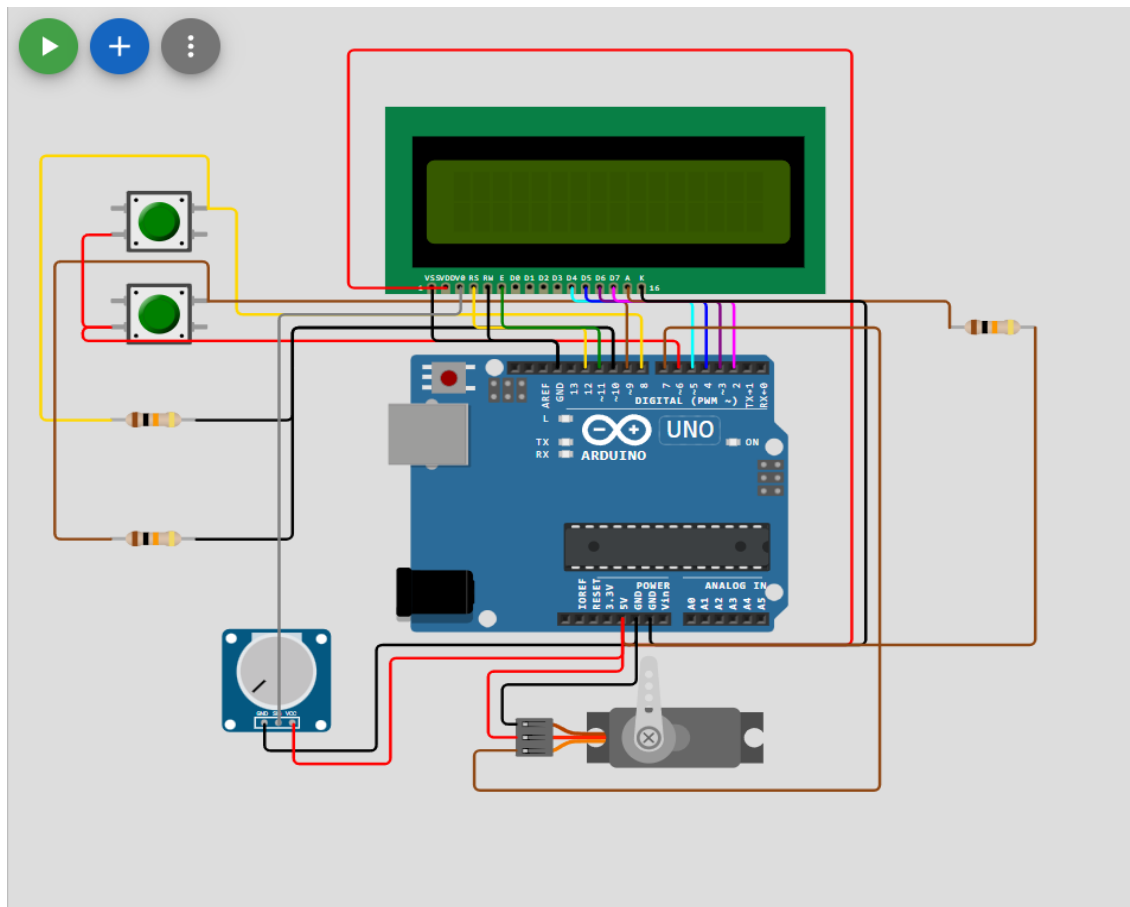
Components used:

1. Arduino UNO
2. Ultrasonic sensor (HC-SR04)
3. Servo motor
4. LCD display (16x2)
5. Power supply
6. Connecting wires
7. Bread board

Block diagram:



Circuit diagram:



Working:

➤ Ultrasonic sensor:

- ❖ Ultrasonic sensor is used to measure the distance between the vehicle and obstacles.
- ❖ It send ultrasonic waves and receives the echo signal. Based on the time taken, distance is calculated and send to Arduino.



➤ Arduino:

- ❖ Arduino acts as the brain of the system. It receives distance data from the sensor and processes it.

- ❖ Based on the program, it decides whether to move forward, stop or start parking.



➤ **Servo motor:**

- ❖ Servo motor is use for steering control.
- ❖ It rotates to a specific angle (left/right) based on Arduino command.
- ❖ This helps in turning the vehicle during parking.



➤ LCD Display:

- ❖ LCD display shows real-time information to the user.
- ❖ It displays messages like:
 - “searching”
 - “parking”
 - “parked”



Coding:

```
#include <Servo.h>
#include <LiquidCrystal.h>

Servo myServo;
LiquidCrystal lcd(7, 8, 11, 12, 13, A0);

int trigPin = 9;
int echoPin = 10;
int motor1 = 2;
int motor2 = 3;

long duration;
int distance;
```

```
void setup() {  
    pinMode(trigPin, OUTPUT);  
    pinMode(echoPin, INPUT);  
    pinMode(motor1, OUTPUT);  
    pinMode(motor2, OUTPUT);  
  
    myServo.attach(6);  
  
    lcd.begin(16, 2);  
    lcd.print("Parking System");  
  
    Serial.begin(9600);  
}
```

```
void loop() {  
  
    lcd.setCursor(0, 1);  
    lcd.print("Searching...");  
  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);  
}
```

```
duration = pulseIn(echoPin, HIGH);  
distance = duration * 0.034 / 2;
```

```
Serial.println(distance);
```

```
if (distance > 20) {  
    digitalWrite(motor1, HIGH);  
    digitalWrite(motor2, LOW);  
    myServo.write(90);  
} else {  
    lcd.clear();  
    lcd.print("Parking...");
```

```
    digitalWrite(motor1, LOW);  
    digitalWrite(motor2, LOW);  
    delay(1000);
```

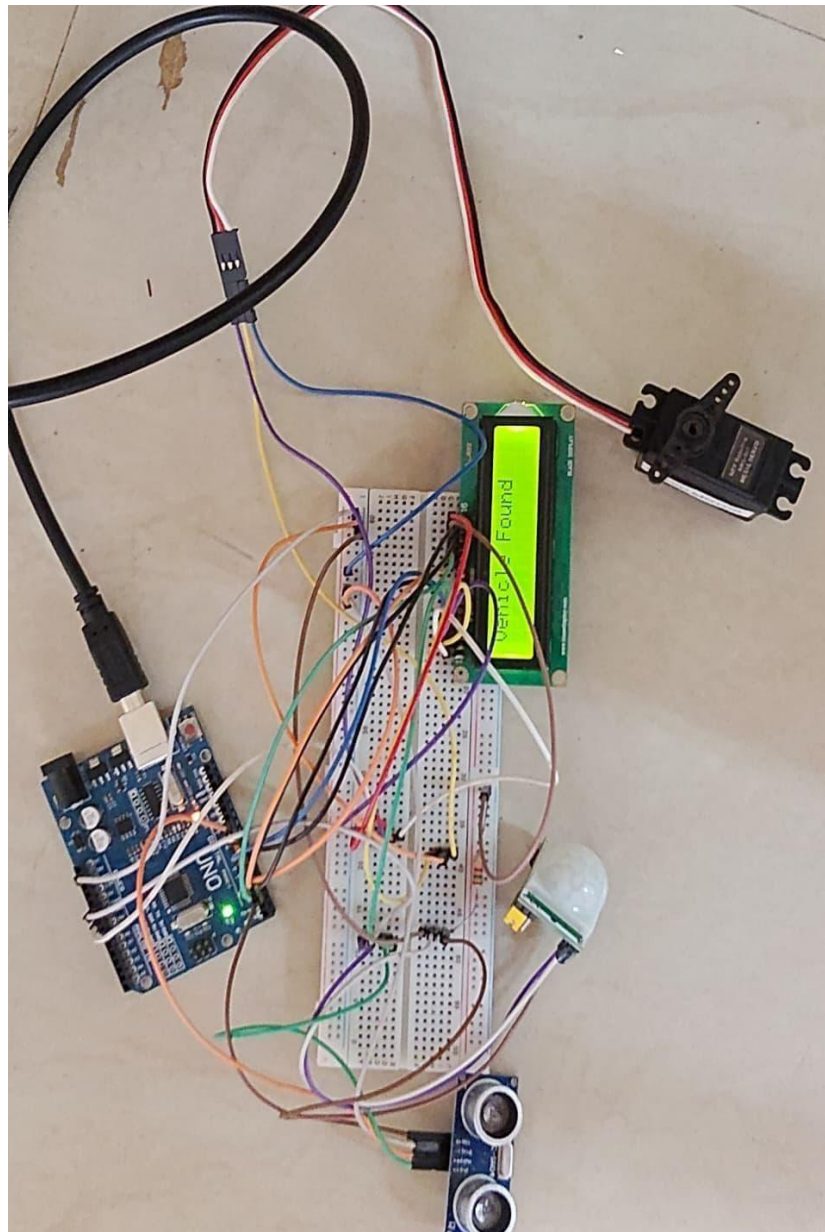
```
    digitalWrite(motor1, LOW);  
    digitalWrite(motor2, HIGH);  
    myServo.write(45);  
    delay(2000);
```

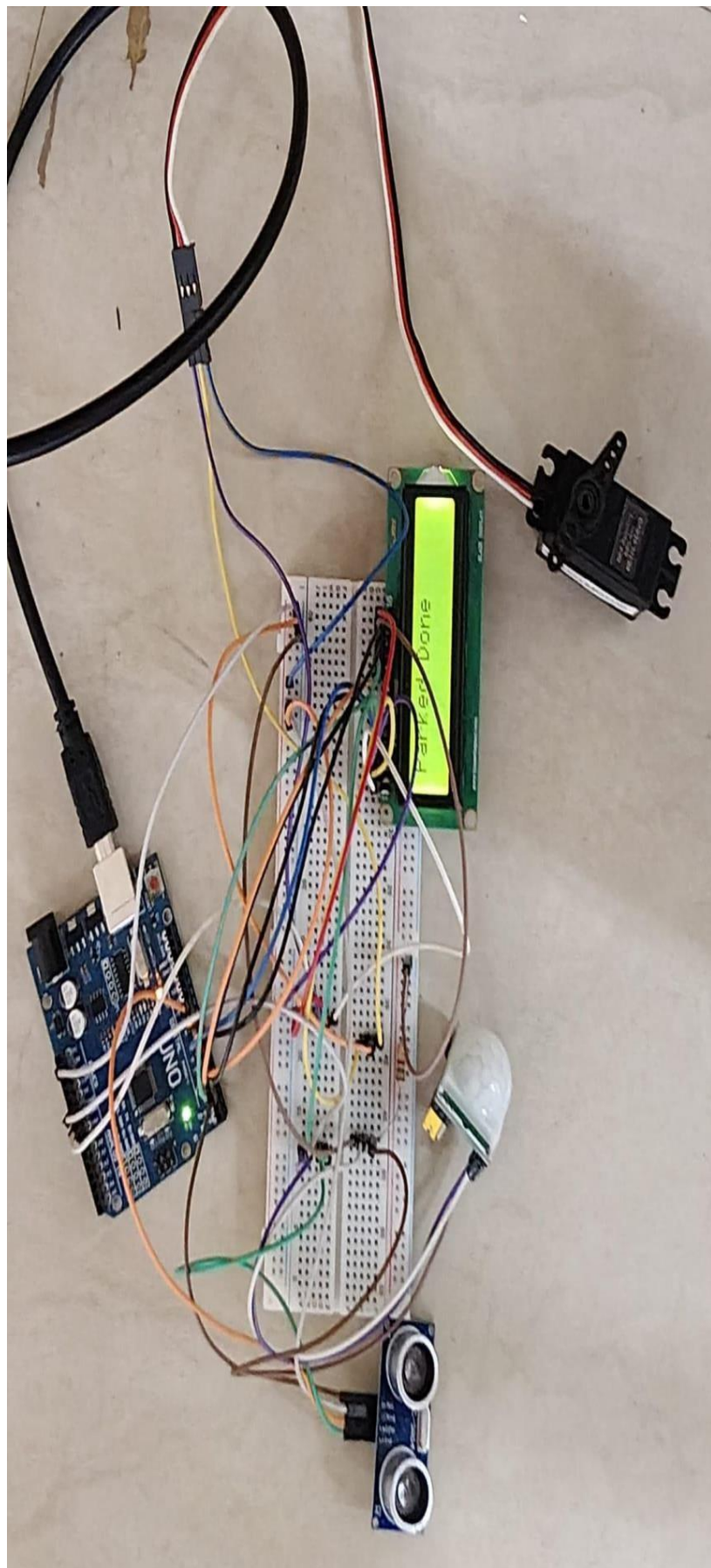
```
    lcd.clear();  
    lcd.print("Parked");
```

```
    digitalWrite(motor1, LOW);
```

```
digitalWrite(motor2, LOW);  
}  
  
delay(500);  
}
```

Output:





Circuit Connections:

➤ Ultrasonic Sensor:

VCC → 5V

GND → GND

TRIG → Pin 9

ECHO → Pin 10

➤ Servo Motor:

VCC → 5V

GND → GND

Signal → Pin 6

➤ LCD (16x2):

RS → Pin 7

E → Pin 8

D4 → Pin 11

D5 → Pin 12

D6 → Pin 13

D7 → A0

VCC → 5V

GND → GND

Advantages:

1. Reduces driver effort.
2. Avoids human errors during parking.
3. Provide accurate parking in tight spaces.
4. Improves safety and prevents collisions.
5. Easy to use and user-friendly.
6. Saves time during parking.
7. Helpful for beginners and inexperienced drivers.
8. Real-time status display using lcd.
9. Low cost and simple implementation.

Conclusion:

- ❖ Autonomous Parallel Parking is an innovative advancement in modern vehicle technology that enhances convenience, safety, and efficiency in parking.
- ❖ By using sensors, cameras, and intelligent control systems, the vehicle can accurately detect suitable

parking spaces and perform parking with minimal or no human intervention.

- ❖ This system significantly reduces driver stress, minimizes the risk of accidents, and ensures precise parking even in tight urban environments.
- ❖ Overall, autonomous parking systems represent an important step toward fully automated vehicles and smarter transportation solutions in the future.

Reference:

1. IoT based Smart Parking System, IEEE ICICCS 2021, Yash Agarwal, Umang Shah, Puru Jain.
2. IoT-Based Shared Community Transportation System Using eBikes, IEEE 2021, A. R. Al-Ali, Raafat Aburukba.
3. An IoT based Smart Outdoor Parking System, ICACCS 2021, Gokul Krishna. S, Harsheetha.